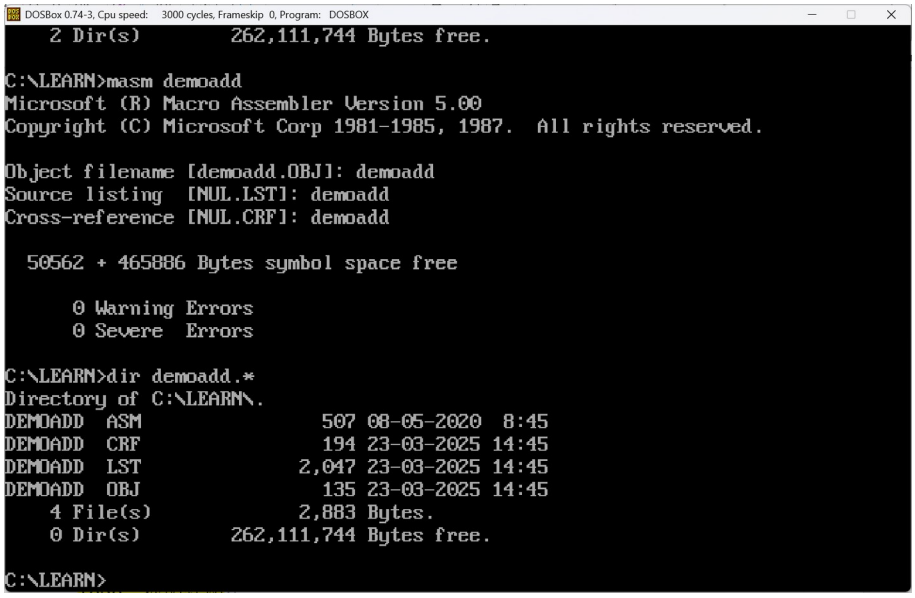
1

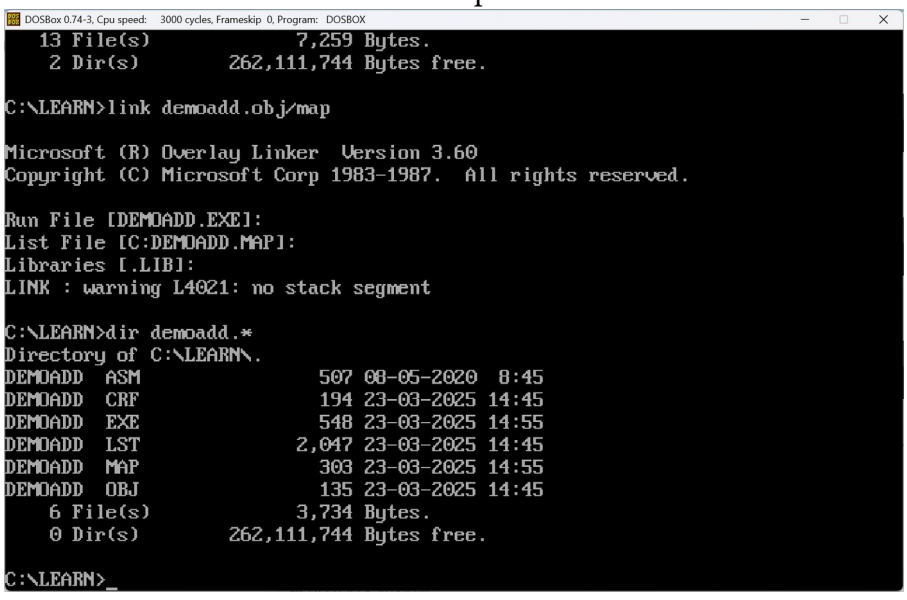
	8	

1 masm link debug

1 masm demoadd.asm .lst .crf



2 link demoadd.asm .map



3 debug demoadd.exe

a U

```
C:\LEARN>debug demoadd.exe
-u
0777:0000 B87607        MOV     AX,0776
0777:0003 8ED8          MOV     DS,AX
0777:0005 A00000        MOV     AL,[0000]
0777:0008 02060100      ADD     AL,[0001]
0777:000C A20200        MOV     [0002],AL
0777:000F B8004C        MOV     AX,4C00
0777:0012 CD21          INT     21
0777:0014 FF00          INC     WORD PTR [BX+SI]
0777:0016 C682FBFE00    MOV     BYTE PTR [BP+SI+FEFB],00
0777:001B 2BC0          SUB     AX,AX
0777:001D 50            PUSH    AX
0777:001E 8D86FBFE      LEA     AX,[BP+FEFB]
-_▲
```

b R

```
-r
AX=FFFF  BX=0000  CX=0024  DX=0000  SP=0000  BP=0000  SI=0000  DI=0000
DS=0766  ES=0766  SS=0775  CS=0777  IP=0000   NV UP EI PL NZ NA PO NC
0777:0000 B87607        MOV     AX,0776
```

c T, 2 T

```
-t

AX=0776  BX=0000  CX=0024  DX=0000  SP=0000  BP=0000  SI=0000  DI=0000
DS=0766  ES=0766  SS=0775  CS=0777  IP=0003   NV UP EI PL NZ NA PO NC
0777:0003 8ED8          MOV     DS,AX
-t

AX=0776  BX=0000  CX=0024  DX=0000  SP=0000  BP=0000  SI=0000  DI=0000
DS=0776  ES=0766  SS=0775  CS=0777  IP=0005   NV UP EI PL NZ NA PO NC
0777:0005 A00000        MOV     AL,[0000]                       DS:0000=11
```

d D 3

```
-d 0 2
0776:0000  11 22 00
```

e G add al, Y

```
-g 0008

AX=0711  BX=0000  CX=0024  DX=0000  SP=0000  BP=0000  SI=0000  DI=0000
DS=0776  ES=0766  SS=0775  CS=0777  IP=0008   NV UP EI PL NZ NA PO NC
0777:0008 02060100      ADD     AL,[0001]                       DS:0001=22
```

f G

```
-g

Program terminated normally
```

2 stu X,Y,Z,W stu X,Y,W

stu

1

```
C:\LEARN>masm work1.asm
Microsoft (R) Macro Assembler Version 5.00
Copyright (C) Microsoft Corp 1981-1985, 1987.  All rights reserved.

Object filename [work1.OBJ]:
Source listing  [NUL.LST]:
Cross-reference [NUL.CRF]:

  51666 + 464782 Bytes symbol space free

      0 Warning Errors
      0 Severe  Errors

C:\LEARN>link work1.obj

Microsoft (R) Overlay Linker  Version 3.60
Copyright (C) Microsoft Corp 1983-1987.  All rights reserved.

Run File [WORK1.EXE]:
List File [NUL.MAP]:
Libraries [.LIB]:
LINK : warning L4021: no stack segment
C:\LEARN>dir work1.*
Directory of C:\LEARN\.
WORK1    ASM               775 23-03-2025 15:33
WORK1    CRF               251 23-03-2025 15:34
WORK1    EXE               562 23-03-2025 15:35
WORK1    LST             2,754 23-03-2025 15:34
WORK1    OBJ               168 23-03-2025 15:35
    5 File(s)          4,510 Bytes.
    0 Dir(s)       262,111,744 Bytes free.
```

2

```
-u
077F:0000 B87E07        MOV     AX,077E
077F:0003 8ED8          MOV     DS,AX
077F:0005 A10E00        MOV     AX,[000E]
077F:0008 2A060B00      SUB     AL,[000B]
077F:000C B305          MOV     BL,05
077F:000E F6F3          DIV     BL
077F:0010 8A1E0C00      MOV     BL,[000C]
077F:0014 F6E3          MUL     BL
077F:0016 B302          MOV     BL,02
077F:0018 B400          MOV     AH,00
077F:001A F6F3          DIV     BL
077F:001C A20D00        MOV     [000D],AL
077F:001F B8004C        MOV     AX,4C00
```

3 stu X Y Z W 5

4 stu X Y Z W 5

```
-d 0 E
0776:0000  32 32 33 36 31 31 34 39-36 38 00 12 34 00 56     2236114968..4.V
```

```
Program terminated normally
-d 0 E
0776:0000  32 32 33 36 31 31 34 39-36 38 00 12 34 52 56      2236114968..4RV
```

5

```
; empty asm file

title I love asm

data segment
    stu   db   '2236114968', 0
    X     db   12H
    Y     db   34H
    Z     db   ?
    W     db   56H
data ends

code segment
    assume cs:code, ds:data
    main      proc
        ; assign the data segment base address to DS
        mov    ax, data
        mov    ds, ax

        ; TODO ...
        ; | add your code between arrows |
        ; v ---------------------------- v
        mov      ax, W
        sub    al, X
        mov    bl, 5
        div    bl
        mov    bl, Y
        mul    bl
        mov    bl, 2
        mov    ah, 0
        div    bl
        mov    Z, al
        ; ^ ---------------------------- ^
        ; |              The END                |

        ; method 2: return to dos
        mov    ax, 4c00h
        int    21h
    main      endp
code ends
```

end main

3 BL CL

1

```
-u
077E:0000 B87E07        MOV     AX,077E
077E:0003 8ED8          MOV     DS,AX
077E:0005 B366          MOV     BL,66
077E:0007 B103          MOV     CL,03
077E:0009 B80000        MOV     AX,0000
077E:000C 8AC3          MOV     AL,BL
077E:000E F6F1          DIV     CL
077E:0010 B102          MOV     CL,02
077E:0012 F6E1          MUL     CL
077E:0014 8BD0          MOV     DX,AX
077E:0016 B8004C        MOV     AX,4C00
077E:0019 CD21          INT     21
077E:001B 4E            DEC     SI
077E:001C 42            INC     DX
077E:001D 3030          XOR     [BX+SI],DH
077E:001F 7000          JO      0021
```

2

```
-t

AX=0000  BX=0066  CX=0003  DX=0000  SP=0000  BP=0000  SI=0000  DI=0000
DS=077E  ES=076E  SS=077D  CS=077E  IP=000C   NV UP EI PL NZ NA PO NC
077E:000C 8AC3          MOV     AL,BL
```

3

```
-t

AX=0044  BX=0066  CX=0002  DX=0044  SP=0000  BP=0000  SI=0000  DI=0000
DS=077E  ES=076E  SS=077D  CS=077E  IP=0016   NV UP EI PL NZ NA PO NC
077E:0016 B8004C        MOV     AX,4C00
```

4

```
    ; empty asm file

title I love asm
data segment

data ends

code segment
    assume cs:code, ds:data
    main    proc
        ; assign the data segment base address to DS
        mov     ax, data
        mov     ds, ax

        ; TODO ...
```

```
        ; | add your code between arrows |
        ; v ---------------------------- v
          mov     bl, 66h
        mov     cl, 03h
        mov     ax, 00h
        mov     al, bl
        div     cl
        mov     cl, 02h
        mul     cl
        mov     dx, al

          ; ^ ---------------------------- ^
        ; |               The END                  |

          ; method 2: return to dos
          mov     ax, 4c00h
          int     21h
    main      endp
code ends
end main
```